Exploring the Data

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NOTE: These tables aren't perfect/are drafts and were done to brainstorm how to frame our findings.

```
# Reading in the data
data <- read_dta("C:/Users/chris/Downloads/covadj_review_final.dta")
head(data)
```

```
# A tibble: 6 x 183
  record_id redcap_survey_identifier date_form  covidenceid reviewername
      <dbl>                    <dbl> <date>           <dbl> <dbl+lbl>
1       334                       NA 2025-10-21           9  5 [Zada Tiimob]
2        71                       NA 2025-08-14          19  4 [Christine Bui]
3       538                       NA 2026-01-21          21  3 [Alyssa Platt]
4       512                       NA 2026-01-12          24 14 [Jack Hall]
5       409                       NA 2025-11-20          31  1 [John Gallis]
6       395                       NA 2025-11-15          38 14 [Jack Hall]
# i 178 more variables: reviewernameoth <dbl>, protocolupload1 <chr>,
#   protocolupload2 <chr>, numpart <dbl>, numunclear <chr>,
#   rzvarsyesno <dbl+lbl>, rzvariables <chr>, studytype <dbl+lbl>,
#   studytypeoth <chr>, primaryoutcomeident <dbl+lbl>, primaryoutcomenum <dbl>,
#   primaryoutcome1 <chr>, primaryoutcome2 <chr>, primoutcometype <dbl+lbl>,
#   primoutcometypeoth <chr>, scalescore <dbl+lbl>, baseoutcome <dbl+lbl>,
#   baseoutcomenowhy___0 <dbl+lbl>, baseoutcomenowhy___1 <dbl+lbl>, ...
```

```
# Create a cleaned version with recoded values
data_cleaned <- data %>%
  mutate(
    # Convert labelled variables to factors with proper labels
    rzvarsyesno_clean = case_when(
```

```
    rzvarsyesno == 1 ~ "Yes",
    rzvarsyesno == 0 ~ "No",
    TRUE ~ "Unclear"
  ),

  # Baseline outcome reported
  baseoutcome_clean = case_when(
    baseoutcome == 1 ~ "Yes",
    baseoutcome == 0 ~ "No",
    TRUE ~ "Unclear"
  ),

  # Adjusted for baseline outcome
  baseoutcomeadj_clean = case_when(
    baseoutcomeadj == 1 ~ "Yes",
    baseoutcomeadj == 0 ~ "No",
    TRUE ~ "Unclear/NA"
  ),

  # Any adjusted analysis
  adjcovar_clean = case_when(
    adjcovar == 1 ~ "Yes",
    adjcovar == 0 ~ "No",
    TRUE ~ "Unclear"
  ),

  # Pre-specification of covariates
  adjprespec_clean = case_when(
    adjprespec == 1 ~ "Pre-specified a priori",
    adjprespec == 0 ~ "Post-hoc",
    adjprespec == -999 ~ "Not reported/Unclear",
    TRUE ~ "Not reported/Unclear"
  ),

  # Journal impact factor category (you'll need to define this based on your data)
  # For now, I'll create a placeholder - adjust based on your actual journal impact data
  high_impact = case_when(
    journal %in% c("The lancet. HIV", "Journal of the American Medical Association",
                   "The Lancet", "New England Journal of Medicine", "BMJ") ~ "High impact
    TRUE ~ "Other journals"
  ),
```

```
    # Number of clusters category
    cluster_size_category = case_when(
      numberofclustersrandomized <= 10 ~ "Small ( 10 clusters)",
      numberofclustersrandomized <= 50 ~ "Medium (11-50 clusters)",
      numberofclustersrandomized > 50 ~ "Large (>50 clusters)",
      TRUE ~ "Not reported"
    )
  )
```

```
# Table 1: Trial Characteristics
table1 <- data_cleaned %>%
  summarise(
    `Total trials` = as.character(n()),
    `Publication year range` = paste(min(publishedyear, na.rm = TRUE),
                                     "-",
                                     max(publishedyear, na.rm = TRUE)),
    `Median publication year` = as.character(median(publishedyear, na.rm = TRUE)),
    `High impact journals` = as.character(sum(high_impact == "High impact (IF > 10)", na.rm =
    `High impact journals (%)` = percent(mean(high_impact == "High impact (IF > 10)", na.rm =
    `Median clusters per trial` = as.character(median(numberofclustersrandomized, na.rm = TRU
    `Range of clusters` = paste(min(numberofclustersrandomized, na.rm = TRUE),
                                "-",
                                max(numberofclustersrandomized, na.rm = TRUE)),
    `Trials with primary outcome identified` = as.character(sum(primaryoutcomeident == 1, na
    `Trials with primary outcome identified (%)` = percent(mean(primaryoutcomeident == 1, na
  ) %>%
  pivot_longer(everything(), names_to = "Characteristic", values_to = "Value")

# Create gt table
table1_gt <- table1 %>%
  gt() %>%
  tab_header(
    title = "Table 1. Characteristics of Included Cluster Randomized Trials",
    subtitle = paste("N =", nrow(data_cleaned), "trials")
  ) %>%
  cols_label(
    Characteristic = "Characteristic",
    Value = "Value"
  ) %>%
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_column_labels(everything())
```

Table 1. Characteristics of Included Cluster Randomized Trials

N = 290 trials

Characteristic	Value
Total trials	290
Publication year range	2017 - 2022
Median publication year	2019
High impact journals	7
High impact journals (%)	2.4%
Median clusters per trial	23
Range of clusters	2 - 999
Trials with primary outcome identified	250
Trials with primary outcome identified (%)	86.2%

```
  ) %>%
  tab_options(
    table.font.size = "small",
    heading.title.font.size = "medium",
    heading.subtitle.font.size = "small"
  )

# Print table
table1_gt
```

Why this table matters for covariate adjustment: (1) publication year range shows whether we're capturing recent practice, (2) journal impact factor could matter a lot (maybe high impact journals are more reliable), (3) primary outcome identification matters because we need to know if trials clearly specified outcomes to assess whether adjustment was pre-specified.

It answers a question: is our sample representative, and do characteristics that might influence covariate handling across trials?

```
# Table 2: Use of covariates in randomization
# Create denominator for location and size
trials_with_location <- data_cleaned %>%
  filter(!is.na(q4selectcountryorcountriesinwhi) & q4selectcountryorcountriesinwhi != "")

trials_with_variable_size <- data_cleaned %>%
  filter(!is.na(numberofclustersrandomized) & numberofclustersrandomized != 999)

# Create the table data
```

```r
table2_data <- data.frame(
  review_item = c(
    "Any covariate used in randomization",
    "Type of covariate used (excluding location and size):",
    "  - Cluster-level only",
    "  - Individual-level only",
    "  - Both cluster- and individual-level",
    "  - None",
    "  - Unclear",
    "Covariates used in randomization:",
    "  - Cluster location",
    "  - Cluster size",
    "Number of covariates used (when any used):",
    "  - 1 covariate",
    "  - 2 covariates",
    "  - 3 or more covariates"
  ),

  n = c(
    sum(data_cleaned$rzvarsyesno == 1, na.rm = TRUE),
    NA,
    sum(data_cleaned$adjcovartype___1 == 1 &
          (data_cleaned$adjcovartype___2 != 1 | is.na(data_cleaned$adjcovartype___2)),
        na.rm = TRUE),
    sum(data_cleaned$adjcovartype___2 == 1 &
          (data_cleaned$adjcovartype___1 != 1 | is.na(data_cleaned$adjcovartype___1)),
        na.rm = TRUE),
    sum(data_cleaned$adjcovartype___1 == 1 & data_cleaned$adjcovartype___2 == 1,
        na.rm = TRUE),
    sum(data_cleaned$rzvarsyesno == 0, na.rm = TRUE),
    sum(is.na(data_cleaned$rzvarsyesno) | data_cleaned$rzvarsyesno == -999, na.rm = TRUE),
    NA,
    sum(str_detect(data_cleaned$rzvariables, "location"), na.rm = TRUE),
    sum(str_detect(data_cleaned$rzvariables, "size"), na.rm = TRUE),
    NA,
    sum(data_cleaned$rzvarsyesno == 1 & data_cleaned$rzvarsnum == 1, na.rm = TRUE),
    sum(data_cleaned$rzvarsyesno == 1 & data_cleaned$rzvarsnum == 2, na.rm = TRUE),
    sum(data_cleaned$rzvarsyesno == 1 & data_cleaned$rzvarsnum >= 3, na.rm = TRUE)
  ),

  stringsAsFactors = FALSE
) %>%
```

```
  mutate(
    N = case_when(
      review_item == "Any covariate used in randomization" ~ nrow(data_cleaned),
      review_item %in% c("  - Cluster-level only", "  - Individual-level only",
                         "  - Both cluster- and individual-level", "  - None", "  - Unclear"
      review_item == "  - Cluster location" ~ nrow(trials_with_location),
      review_item == "  - Cluster size" ~ nrow(trials_with_variable_size),
      review_item %in% c("  - 1 covariate", "  - 2 covariates", "  - 3 or more covariates") ~
        sum(data_cleaned$rzvarsyesno == 1, na.rm = TRUE),
      TRUE ~ NA_real_
    ),
    `%` = if_else(is.na(N) | is.na(n), "", sprintf("%.1f%%", (n / N) * 100)),
    `n/N` = if_else(is.na(N) | is.na(n), "", paste0(n, "/", N))
  ) %>%
  select(review_item, `n/N`, `%`)

# Create gt table
table2_gt <- table2_data %>%
  gt() %>%
  tab_header(
    title = "Table 2. Use of Covariates in Randomization",
    subtitle = paste("Based on", nrow(data_cleaned), "cluster randomized trials")
  ) %>%
  cols_label(
    review_item = "Review item",
    `n/N` = "n/N",
    `%` = "%"
  ) %>%
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_column_labels(everything())
  ) %>%
  tab_style(
    style = cell_text(style = "italic"),
    locations = cells_body(rows = c(2, 8, 11))
  ) %>%
  tab_options(
    table.font.size = "small",
    heading.title.font.size = "medium"
  )

table2_gt
```

Table 2. Use of Covariates in Randomization

Based on 290 cluster randomized trials

Review item	**n/N**	%
Any covariate used in randomization	160/290	55.2%
Type of covariate used (excluding location and size):		
- Cluster-level only	19/290	6.6%
- Individual-level only	121/290	41.7%
- Both cluster- and individual-level	47/290	16.2%
- None	117/290	40.3%
- Unclear	13/290	4.5%
Covariates used in randomization:		
- Cluster location	16/290	5.5%
- Cluster size	32/288	11.1%
Number of covariates used (when any used):		
- 1 covariate	0/160	0.0%
- 2 covariates	0/160	0.0%
- 3 or more covariates	0/160	0.0%

This table assesses whether researchers are using restricted randomization to promote baseline balance. It highlights type of covariates used, fundamental cluster characteristics (location and size), and number of covariates.

```
# Table 3: Reporting of baseline covariates
# Create the summary data using existing cleaned variables
table3_data <- data.frame(
  review_item = c(
    "Baseline table includes:",
    "  - Cluster-level covariates only",
    "  - Individual-level covariates only",
    "  - Both cluster- and individual-level",
    "  - No table",
    "Baseline measure of primary outcome reported",
    "Significance test of balance reported"
  ),

  n = c(
    NA,  # Section header
    sum(data_cleaned$covreportedarm___1 == 1 &
          (data_cleaned$covreportedarm___2 != 1 | is.na(data_cleaned$covreportedarm___2)),
        na.rm = TRUE),
    sum(data_cleaned$covreportedarm___2 == 1 &
```

```
          (data_cleaned$covreportedarm___1 != 1 | is.na(data_cleaned$covreportedarm___1)),
        na.rm = TRUE),
    sum(data_cleaned$covreportedarm___1 == 1 & data_cleaned$covreportedarm___2 == 1,
        na.rm = TRUE),
    sum(is.na(data_cleaned$covreported) | data_cleaned$covreported == 0,
        na.rm = TRUE),
    sum(data_cleaned$baseoutcome_clean == "Yes", na.rm = TRUE),
    sum(data_cleaned$baseoutcomenowhy___555 == 555, na.rm = TRUE)
  ),

  stringsAsFactors = FALSE
) %>%
  mutate(
    N = case_when(
      review_item %in% c("Baseline table includes:", "  - Cluster-level covariates only",
                         "  - Individual-level covariates only", "  - Both cluster- and indi
                         "  - No table") ~ nrow(data_cleaned),
      review_item == "Baseline measure of primary outcome reported" ~ nrow(data_cleaned),
      review_item == "Significance test of balance reported" ~ nrow(data_cleaned),
      TRUE ~ NA_real_
    ),
    `%` = if_else(is.na(N) | is.na(n), "", sprintf("%.1f%%", (n / N) * 100)),
    `n/N` = if_else(is.na(N) | is.na(n), "", paste0(n, "/", N))
  ) %>%
  select(review_item, `n/N`, `%`)

# Create gt table
table3_gt <- table3_data %>%
  gt() %>%
  tab_header(
    title = "Table 3. Reporting of Baseline Covariates",
    subtitle = paste("Based on", nrow(data_cleaned), "cluster randomized trials")
  ) %>%
  cols_label(
    review_item = "Review item",
    `n/N` = "n/N",
    `%` = "%"
  ) %>%
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_column_labels(everything())
  ) %>%
```

Table 3. Reporting of Baseline Covariates
Based on 290 cluster randomized trials

Review item	**n/N**	%
Baseline table includes:		
- Cluster-level covariates only	2/290	0.7%
- Individual-level covariates only	108/290	37.2%
- Both cluster- and individual-level	38/290	13.1%
- No table	142/290	49.0%
Baseline measure of primary outcome reported	210/290	72.4%
Significance test of balance reported	0/290	0.0%

```
  tab_style(
    style = cell_text(style = "italic"),
    locations = cells_body(rows = 1)
  ) %>%
  tab_options(
    table.font.size = "small",
    heading.title.font.size = "medium"
  )

# Display the table
table3_gt
```

This table evaluates transparency in reporting what was actually balanced or imbalanced. It answers the question: are trials transparent about baseline comparability?

```
# Table 4: Covariate adjustment in analysis
# Create denominators
trials_with_baseoutcome <- data_cleaned %>% filter(baseoutcome == 1)
trials_with_rzvars <- data_cleaned %>% filter(rzvarsyesno == 1)
trials_with_adj <- data_cleaned %>% filter(adjcovar == 1)

# Create the summary data
table4_data <- data.frame(
  review_item = c(
    "Trial report includes an adjusted analysis",
    "",
    "Adjusted for baseline measure of outcome:",
    "  - Among trials reporting baseline outcome",
    "Adjusted for all covariates used in randomization:",
```

```
    "  - Among trials using covariates in randomization",
    "Timing of covariate selection:",
    "  - Pre-specified a priori",
    "  - Post-hoc",
    "  - Unclear/not reported",
    "Level of covariates in adjusted analysis:",
    "  - Cluster-level only",
    "  - Individual-level only",
    "  - Both cluster- and individual-level",
    "  - Unclear"
  ),

  n = c(
    sum(data_cleaned$adjcovar == 1, na.rm = TRUE),
    NA,
    NA,
    sum(trials_with_baseoutcome$baseoutcomeadj == 1, na.rm = TRUE),
    NA,
    sum(trials_with_rzvars$rzvarsused == 1, na.rm = TRUE),
    NA,
    sum(data_cleaned$adjprespec == 1, na.rm = TRUE),
    sum(data_cleaned$adjprespec == 0, na.rm = TRUE),
    sum(data_cleaned$adjprespec == -999 | is.na(data_cleaned$adjprespec), na.rm = TRUE),
    NA,
    sum(data_cleaned$adjcovartype___1 == 1 &
          (data_cleaned$adjcovartype___2 != 1 | is.na(data_cleaned$adjcovartype___2)),
        na.rm = TRUE),
    sum(data_cleaned$adjcovartype___2 == 1 &
          (data_cleaned$adjcovartype___1 != 1 | is.na(data_cleaned$adjcovartype___1)),
        na.rm = TRUE),
    sum(data_cleaned$adjcovartype___1 == 1 & data_cleaned$adjcovartype___2 == 1,
        na.rm = TRUE),
    sum((data_cleaned$adjcovartype___1 == 0 | is.na(data_cleaned$adjcovartype___1)) &
          (data_cleaned$adjcovartype___2 == 0 | is.na(data_cleaned$adjcovartype___2)),
        na.rm = TRUE)
  ),

  stringsAsFactors = FALSE
) %>%
  mutate(
    N = case_when(
      review_item == "Trial report includes an adjusted analysis" ~ nrow(data_cleaned),
```

```
      review_item == "  - Among trials reporting baseline outcome" ~ nrow(trials_with_baseout
      review_item == "  - Among trials using covariates in randomization" ~ nrow(trials_with_
      review_item %in% c("  - Pre-specified a priori", "  - Post-hoc", "  - Unclear/not repor
      review_item %in% c("  - Cluster-level only", "  - Individual-level only",
                         "  - Both cluster- and individual-level", "  - Unclear") ~ nrow(tri
      TRUE ~ NA_real_
    ),
    `%` = if_else(is.na(N) | is.na(n), "", sprintf("%.1f%%", (n / N) * 100)),
    `n/N` = if_else(is.na(N) | is.na(n), "", paste0(n, "/", N))
  ) %>%
  select(review_item, `n/N`, `%`)

# Create gt table
table4_gt <- table4_data %>%
  gt() %>%
  tab_header(
    title = "Table 4. Covariate Adjustment in Analysis of Primary Outcome",
    subtitle = paste("Based on", nrow(data_cleaned), "cluster randomized trials")
  ) %>%
  cols_label(
    review_item = "Review item",
    `n/N` = "n/N",
    `%` = "%"
  ) %>%
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_column_labels(everything())
  ) %>%
  tab_style(
    style = cell_text(style = "italic"),
    locations = cells_body(rows = c(2, 4, 6, 11))
  ) %>%
  tab_options(
    table.font.size = "small",
    heading.title.font.size = "medium"
  )

table4_gt
```

This table assesses whether and how covariates are incorporated into the primary analysis. It asks a question: are researchers adjusting for covariates appropriately, and are they transparent about how they chose which covariates to include?

Table 4. Covariate Adjustment in Analysis of Primary Outcome
Based on 290 cluster randomized trials

Review item	**n/N**	%
Trial report includes an adjusted analysis	187/290	64.5%
Adjusted for baseline measure of outcome:		
- Among trials reporting baseline outcome	*158/210*	*75.2%*
Adjusted for all covariates used in randomization:		
- Among trials using covariates in randomization	*48/160*	*30.0%*
Timing of covariate selection:		
- Pre-specified a priori	82/187	43.9%
- Post-hoc	42/187	22.5%
- Unclear/not reported	166/187	88.8%
Level of covariates in adjusted analysis:		
- Cluster-level only	19/187	10.2%
- Individual-level only	121/187	64.7%
- Both cluster- and individual-level	47/187	25.1%
- Unclear	103/187	55.1%